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FURTHER READING AND RESOURCES



This appendix lists further reading and resources that can serve as great companions to this book. Some offer deeper dives into content merely touched on here, while others supplement the book's discussions or focus exclusively on a particular tool or approach. At the end of this appendix, you'll find a list of online sandboxes and malware sources to help you get started practicing the techniques you've seen throughout this book.

Part I (Chapters 1–3)

Eagle, Chris, *The IDA Pro Book: The Unofficial Guide to the World's Most Popular Disassembler*, 2nd ed. San Francisco: No Starch Press, 2011. This is the ultimate guide to the IDA Pro disassembler and de-

bugger. It covers everything from navigating the IDA Pro interface to using plug-ins and some of the tool's more advanced features.

Eagle, Chris, and Kara Nance. *The Ghidra Book: The Definitive Guide*. San Francisco: No Starch Press, 2020. Eagle and Nance cover the basics of using the Ghidra disassembler as well as the more advanced uses of the tool.

Kleymenov, Alexey, and Amr Thabet. *Mastering Malware Analysis: A Malware Analyst's Practical Guide to Combating Malicious Software, APT, Cybercrime, and IoT Attacks*. Birmingham, UK: Packt, 2022. This is a modern, thorough book on malware analysis concepts and techniques. It even covers areas such as Linux and IoT malware and macOS and iOS threats.

Sikorski, Michael, and Andrew Honig. *Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software*. San Francisco: No Starch Press, 2012. This book was one of the first complete guides to malware analysis. Even though the book is now more than a decade old, many of the techniques discussed throughout it are still very relevant.

Yosifovich, Pavel, Mark E. Russinovich, Alex Ionescu, and David A. Solomon. *Windows Internals, Part 1: System Architecture, Processes, Threads, Memory Management, and More*, 7th ed. Redmond, WA: Microsoft Press, 2017; Allievi, Andrea, Alex Ionescu, David A. Solomon, and Mark E. Russinovich. *Windows Internals, Part 2*, 7th ed. Redmond, WA: Microsoft Press, 2021. Both *Windows Internals* books offer an amazingly thorough breakdown of how Windows works under the hood. If you want to understand the Windows architecture in depth, look no further than these books.

Parts II–IV (Chapters 4–17)

Andriesse, Dennis. *Practical Binary Analysis: Build Your Own Linux Tools for Binary Instrumentation, Analysis, and Disassembly*. San Francisco: No Starch Press, 2018. This book explores techniques for analyzing various types of binaries. It covers some of the concepts

that I've introduced only briefly here, such as binary instrumentation.

Hand, Matt. *Evading EDR: The Definitive Guide to Defeating Endpoint Detection Systems*. San Francisco: No Starch Press, 2023. Hand covers a wealth of information on endpoint detection and response (EDR) systems, including their general architecture and strategies for evading them.

Ligh, Michael Hale, Andrew Case, Jamie Levy, and Aaron Walters. *The Art of Memory Forensics: Detecting Malware and Threats in Windows, Linux, and Mac Memory*. Hoboken, NJ: Wiley, 2014. The authors explain the complex topics of memory and memory forensics—a concept I've mentioned only briefly in this book—in great detail.

Matrosov, Alex, Eugene Rodionov, and Sergey Bratus. *Rootkits and Bootkits: Reversing Modern Malware and Next Generation Threats*. San Francisco: No Starch Press, 2019. This book covers modern rootkits and bootkits in depth, as well as how to investigate them from a malware analysis and forensics perspective.

MITRE ATT&CK (<https://attack.mitre.org>) is a knowledge base of the techniques in use by threat actors and is a great companion for nearly all the topics in this book.

Unprotect (<https://unprotect.it>) is a project maintained by researchers Jean-Pierre Lesueur and Thomas Roccia that seeks to categorize the many techniques of malware evasion. It's another excellent companion resource to this book.

Yason, Mark Vincent. "The Art of Unpacking." Atlanta: IBM Internet Security Systems, 2011. <https://www.blackhat.com/presentations/bh-usa-07/Yason/Whitepaper/bh-usa-07-yason-WP.pdf>. While a bit old now, this research paper contains a wealth of information on unpacking malware. It also delves into topics like anti-analysis and debugger attacks.

Yehoshua, Nir, and Uriel Kosayev. *Antivirus Bypass Techniques: Learn Practical Techniques and Tactics to Combat, Bypass, and Evade Antivirus Software*. Birmingham, UK: Packt, 2021. This book contains specialized knowledge on evading anti-malware defenses.

While primarily targeted to “offensive” researchers, the techniques in the book can help malware analysts understand how malware bypasses these defenses.

Online Malware Sandboxes

This is a list of some of the free (or partly free) sandboxes. You may submit to and analyze malware with these services, but some of them have limited capabilities for their free tier. Also be aware that some of these sandboxes may share data to unknown parties, and the samples may be available to a large audience. Only submit files that do not contain sensitive data, and use at your own risk!

- Any.Run: <https://any.run>
- Cuckoo: <https://cuckoo.cert.ee>
- Hybrid Analysis: <https://hybrid-analysis.com>
- Joe Sandbox: <https://joesandbox.com>
- Triage: <https://tria.ge>
- UnpacMe: <https://unpac.me/>
- VirusTotal: <https://www.virustotal.com>
- Yomi: <https://yomi.yoroi.company/>

Malware Sources

If your day job does not involve investigating new malware, you’ll likely need a source for malware samples you can practice with. The following sources are free (some require a simple sign-up), with the exception of VirusTotal.

- MalShare: <https://malshare.com>
- MalwareBazaar: <https://bazaar.abuse.ch>
- Malware Traffic Analysis: <https://www.malware-traffic-analysis.net>
- VirusShare: <https://virusshare.com>

- VirusSign: <https://www.virussign.com>
- VirusTotal: <https://www.virustotal.com>
- vx-underground: <https://www.vx-underground.org>